<u>Last time</u>: In a metric space (X, d),

- $\{x_n\}$ converges $\Rightarrow$ $\{x_n\}$ Cauchy
- $\{x_n\}$ converges $\not\Leftarrow$ $\{x_n\}$ Cauchy.

  E.g., the sequence

$$x_1 = 3, \qquad x_{n+1} = \frac{x_n}{2} + \frac{3}{2x_n} \quad \forall n \geq 1$$

is Cauchy but does not converge in $(\mathbb{Q}, |\cdot|)$

<u>Def</u> A metric space (X, d) is <u>complete</u> if every Cauchy sequence in X converges in X.

<u>Ex</u> ① $(\mathbb{R}, |\cdot|)$ is complete (by lecture 12)

② $(\mathbb{Q}, |\cdot|)$ is not complete (by the previous example)

<u>Prop</u> Let $1 \leq p \leq \infty$. $\forall n \in \mathbb{N}$, $(\mathbb{R}^n, d_p)$ is complete.

<u>Pf</u>: We'll argue using induction.

Base case: For $n = 1$,

$$d_p(x, y) = \left(|x-y|^p\right)^{\frac{1}{p}} = |x-y|.$$

We know $(\mathbb{R}, |\cdot|)$ is complete by lecture 12.

Inductive step: Assume $(\mathbb{R}^n, d_p)$ is complete for some $n \geq 1$. Want: $(\mathbb{R}^{n+1}, d_p)$ is complete. Let $\{x^k\}_{k \geq 1}$ be Cauchy, and write $x^k = (x_1^k, x_2^k, \ldots, x_{n+1}^k) \in \mathbb{R}^{n+1}$.

Then, $\forall \varepsilon > 0$ $\exists N$ s.t.

$$\underbrace{\left(|x_1^k - x_1^\ell|^p + \ldots + |x_{n+1}^k - x_{n+1}^\ell|^p\right)^{\frac{1}{p}}} = d_p(x^k, x^\ell) < \varepsilon \quad \forall k, \ell \geq N$$

$$\geq \left(|x_1^k - x_1^\ell|^p + \dots + |x_n^k - x_n^\ell|^p\right)^{\frac{1}{p}}$$

So $\tilde{x}^k = (x_1^k, \dots, x_n^k)$ is a Cauchy sequence in $(\mathbb{R}^n, d_p)$. As $(\mathbb{R}^n, d_p)$ is complete, $\exists\, \tilde{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ s.t. $\tilde{x}^k \to \tilde{x}$ in $(\mathbb{R}^n, d_p)$ as $k \to \infty$. Similarly,

$$\left(|x_1^k - x_1^\ell|^p + \dots + |x_{n+1}^k - x_{n+1}^\ell|^p\right)^{\frac{1}{p}} \geq \left(|x_{n+1}^k - x_{n+1}^\ell|^p\right)^{\frac{1}{p}}$$
$$= |x_{n+1}^k - x_{n+1}^\ell|$$

So $\{x_{n+1}^k\}_{k \geq 1}$ is a Cauchy sequence in $(\mathbb{R}, |\cdot|)$. As $(\mathbb{R}, |\cdot|)$ is complete, $\exists\, x_{n+1} \in \mathbb{R}$ s.t. $x_{n+1}^k \to x_{n+1}$ in $(\mathbb{R}, |\cdot|)$ as $k \to \infty$.

Set $x = (\tilde{x}, x_{n+1}) \in \mathbb{R}^{n+1}$. Claim: $x^k \to x$ in $(\mathbb{R}^{n+1}, d_p)$.

$$d_p(x^k, x) = \left(|x_1^k - x_1|^p + \dots + |x_{n+1}^k - x_{n+1}|^p\right)^{\frac{1}{p}}$$
$$= \left(d_p(\tilde{x}^k, \tilde{x})^p + |x_{n+1}^k - x_{n+1}|^p\right)^{\frac{1}{p}}$$
$$\leq 2^{\frac{1}{p}} \cdot \max\{\underbrace{d_p(\tilde{x}^k, \tilde{x})}_{\to 0}, \underbrace{|x_{n+1}^k - x_{n+1}|}_{\to 0}\}$$

Know: $\to 0$ $\to 0$ as $k \to \infty$

So $d_p(x^k, x) \to 0$ as $k \to \infty$ (by the squeeze theorem), and thus $\{x^k\}_{k \geq 1}$ converges to x in $(\mathbb{R}^{n+1}, d_p)$. $\square$

<u>Prop</u> Let (X, d) be a metric space, $A \subseteq X$, and $x \in X$. Then: $x \in \overline{A} \iff \exists$ a sequence $\{a_n\}_{n \geq 1}$ in A

that converges to x.

<u>Rmk</u> Recall: $\bar{A} = A \cup A'$

adherent points limit points

$x \in A' \iff \exists$ a sequence in $A \setminus \{x\}$ that converges to x.

<u>Pf</u>: <u>$\Rightarrow$|</u>: Fix $x \in \bar{A}$. Then $B_r(x) \cap A \neq \emptyset$ $\forall r > 0$. Taking $r = \frac{1}{n}$ for $n \in \mathbb{N}$, we see that $\exists\, a_n$ in $B_{\frac{1}{n}}(x) \cap A$. Then $\{a_n\}_{n \geq 1}$ is a sequence in A, and

$$0 \leq d(a_n, x) < \frac{1}{n} \to 0 \quad \text{as } n \to \infty.$$

Therefore $d(a_n, x) \to 0$ as $n \to \infty$, and so $\{a_n\}_{n \geq 1}$ converges to x.

<u>$\Leftarrow$|</u>: Suppose $\{a_n\}_{n \geq 1}$ is a sequence in A that converges to $x \in X$. Want: $x \in \bar{A}$. Fix $r > 0$. As $\{a_n\}_{n \geq 1}$ converges to x, $\exists\, N$ s.t.

$$\underbrace{d(a_n, x) < r}_{\Rightarrow\ a_n \in B_r(x)} \quad \forall n \geq N.$$

In particular, $a_N \in B_r(x) \cap A$, so $B_r(x) \cap A$ is nonempty. $\square$

Compact sets

<u>Def</u> Let (X, d) be a metric space and $A \subseteq X$. We say:

① a collection of sets $\{\mathcal{U}_i\}_{i\in I}$ is an <u>open cover</u> of A if $I \neq \emptyset$ is a set, $\mathcal{U}_i$ is open $\forall i \in I$, and $A \subseteq \bigcup_{i\in I} \mathcal{U}_i$.

② the set A is <u>compact</u> if: $\forall$ open cover $\{\mathcal{U}_i\}_{i\in I}$ of A, $\exists$ a finite set $J \subseteq I$ s.t. $\{\mathcal{U}_i\}_{i\in J}$ is an open cover of A. We call $\{\mathcal{U}_i\}_{i\in J}$ a <u>finite subcover</u>.

<u>Ex</u> Any finite set $\{x_1, \dots, x_n\} \subseteq X$ is compact:
let $\{\mathcal{U}_i\}_{i\in I}$ be an open cover of $\{x_1, \dots, x_n\}$.
Then

$$x_1 \in \bigcup_{i\in I} \mathcal{U}_i \Rightarrow \exists\, i_1 \in I \text{ s.t. } x_1 \in \mathcal{U}_{i_1}$$

$$x_2 \in \bigcup_{i\in I} \mathcal{U}_i \Rightarrow \exists\, i_2 \in I \text{ s.t. } x_2 \in \mathcal{U}_{i_2}$$

$$\vdots$$

So $\{\mathcal{U}_{i_1}, \dots, \mathcal{U}_{i_n}\}$ is a finite subcover.